

Generative AI and Labor Market Outcomes: Evidence from the United Kingdom

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Abstract

This paper examines the effects of LLMs on UK labor market outcomes. We use a difference-in-differences design that compares outcomes across firms and occupations from 2021 to 2025 based on their differential exposure to LLM capabilities. Our results show that highly exposed firms reduce employment, with effects concentrated in junior positions. These firms sharply curtail new hiring, with technical and creative roles experiencing the steepest declines. At the occupation level, roles more exposed to LLMs also show substantial reductions in job listings. The observed effects concentrate almost entirely in high-wage segments. Hence, our findings suggest that LLMs may compress wage inequality.

Keywords: Generative AI; Large Language Models; Automation; Labor demand; United Kingdom.

JEL: O33; J23; J31

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tal based on comparative advantage. When automation enables capital to perform tasks previously done by labor more cheaply, it creates a *displacement effect* that reduces labor demand. Conversely, the creation of new tasks where labor maintains comparative advantage generates a *reinstatement effect* that increases labor demand (Acemoglu and Restrepo, 2019; Restrepo, 2024). For LLMs, which can automate cognitive tasks previously considered non-routine, the framework predicts that displacement will dominate in the short run, particularly for entry-level positions involving routine information processing and basic analysis. Senior roles requiring tacit knowledge, judgment, and complex interpersonal coordination should prove less vulnerable. This is consistent with our findings.

Our findings contribute to the emerging international evidence on labor market impacts of AI (del Rio-Chanona et al., 2025). In the United States, Brynjolfsson, Chandar and Chen (2025); Lichtinger and Hosseini Maasoum (2025); Dominski and Lee (2025) document employment declines in exposed industries concentrated in early-career positions. However, Chandar (2025) finds no employment effects using CPS data, and Humlum and Vestergaard (2025) report null results from Danish administrative data. These disparities may reflect several factors: differences in labor market institutions between countries, variation in the timing of LLM adoption across regions, and methodological choices regarding exposure measurement and outcome variables. Our UK evidence suggests that effects manifest differently across adjustment margins, with hiring responding more quickly and strongly than employment stocks.

Methodologically, our paper builds on recent work that also uses Revelio Labs data to construct firm-level AI exposure measures. Eisfeldt et al. (2023) show that higher-exposure US firms experience excess stock returns following the release of ChatGPT, while Hampole et al. (2025) develop task-level exposure measures varying across firms and occupations to study US labor demand between 2010 and 2023. We extend their work in three directions: we shift focus to the UK labor market, examine realized employment and job posting outcomes (as opposed to Eisfeldt et al. (2023)’s focus on financial data), and exploit data through mid-2025 to capture labor market adjustment to generative AI adoption (as opposed to Hampole et al. (2025)’s focus on earlier waves of AI).

In the UK context, we extend Henseke et al. (2025), who develop a Generative AI Susceptibility Index and provide suggestive evidence that wage premia decline

sample construction by varying minimum firm size thresholds. We consider alternative task-level weights to construct our exposure measures. We estimate weighted regressions that give greater weight to larger firms and occupations to ensure our results reflect typical worker experiences. Finally, we relax the assumption of linear treatment effects by categorizing firms and occupations into exposure quartiles. These analyses consistently support our main findings.

These robustness checks notwithstanding, we acknowledge the possibility that other factors differentially affected exposed and less exposed firms or occupations around the time of ChatGPT’s introduction. For instance, if firms anticipating AI disruption had already begun adjusting their workforce composition before November 2022, or if other technological or economic shocks coincidentally affected high-exposure entities differently, our estimates could be biased. Additionally, our exposure measures capture potential susceptibility to LLM disruption but may not perfectly predict actual adoption patterns, which likely vary with firm-specific factors such as technological capabilities, management quality, and financial resources.

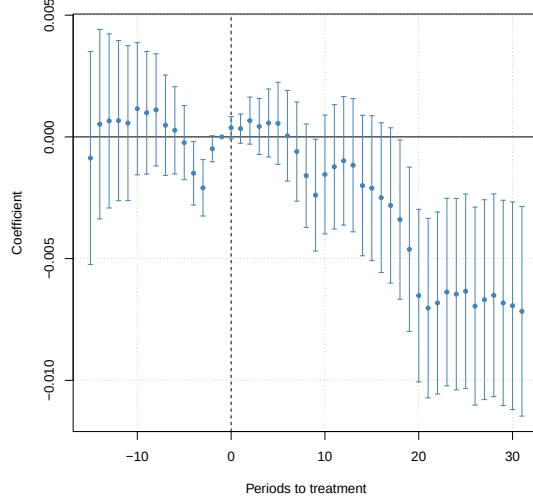
4 Results

This section presents our main empirical findings on the effects of LLM exposure on UK labor market outcomes. We organize the analysis in three parts. First, we examine how employment stocks, workforce composition, and compensation evolve within firms with different levels of LLM exposure. Second, we investigate occupation-level hiring patterns through job posting data. Third, we explore within-firm recruitment strategies by combining firm-level exposure measures with their occupation-specific hiring patterns. Throughout, we present both event study specifications and static difference-in-differences estimates.

4.1 Firm-Level Employment Effects

Figure 1 presents the event study plots for the effect of LLM exposure on the total number of employees in UK firms. We find that highly exposed firms start experiencing a statistically significant decline in total employment beginning approximately 18 months after the introduction of ChatGPT, suggesting that exposed firms reduce headcount in response to this new technology, but that it takes time to

Figure 1: Effect of LLM exposure on firm-level jobs



Notes: This figure plots event study estimates of the effect of LLM exposure on the total number of employees in UK firms (log). Coefficients are estimated from Equation (3). All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. Vertical line indicates November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

show up in aggregate employment statistics. High and low LLM exposure exhibit similar employment trajectories prior to November 2022, which provides support for the validity of the parallel trends assumption. When averaged over the entire post-treatment period (Table 3), a one standard deviation increase in LLM exposure is associated with a 0.3 percent reduction in total employment, although the aggregate effect is only marginally significant due to the delayed onset of impacts.

We next examine how these aggregate effects vary across different types of workers. Figure 2 decomposes the effects between high- and low-seniority employees. Panel A shows junior positions decline immediately and statistically significantly, while Panel B demonstrates that senior positions initially remain stable before declining gradually, with the decrease only becoming significant after 20 months. Panel C shows that the timing difference in the emergence of effects leads to a temporary shift toward senior employees, although the compositional change attenuates over time, with the seniority distribution gradually reverting toward pre-treatment levels by the end of our observation period. All three panels show parallel pre-trends.

Table 3: Effect of LLM exposure on firm-level job outcomes

	Employees	Low seniority	High seniority	Frac. low seniority	Av. comp.
LLM exposure x Post	−0.003* (0.002)	−0.004** (0.002)	−0.002 (0.002)	−0.001** (0.0003)	106.313* (59.029)
Firm fixed effects	Yes	Yes	Yes	Yes	Yes
Industry-by-period FE	Yes	Yes	Yes	Yes	Yes
Observations	1,268,013	1,267,955	1,266,331	1,267,955	1,268,013
Adj. R-squared	0.983	0.983	0.981	0.979	0.978

Notes: This table reports difference-in-differences estimates of the effect of LLM exposure on firm-level employment outcomes, averaged over the post-treatment period. *Employees* is the total number of jobs in the firm. *Low seniority* and *High seniority* denote the number of positions in seniority levels 1–3 and 4–7, respectively. *Low seniority (%)* is the share of jobs that are low-seniority. *Av. comp.* is the mean annual salary plus bonuses. All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level and reported in parentheses. Significance is denoted by * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

The regression results in Table 3 show, averaged over the entire post-treatment period, a significant 0.4 percent reduction in junior employees per standard deviation increase in exposure, and a statistically insignificant reduction in senior employees. The fraction of junior employees goes down by 0.1 percentage points per standard deviation of LLM exposure ($p = 0.028$).

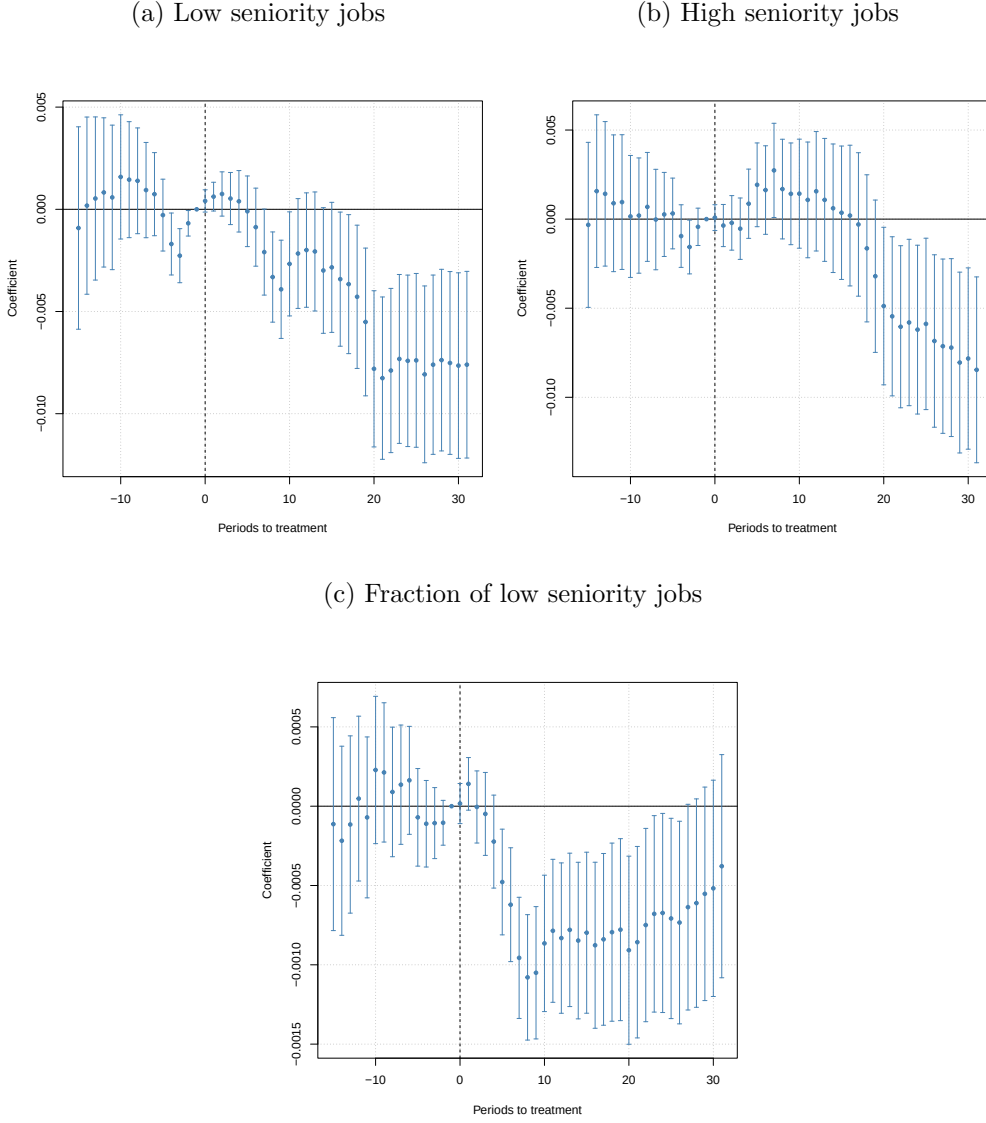
The observed compositional shift towards more senior roles has direct implications for compensation patterns. Figure 3 shows that average compensation rises modestly in highly exposed firms following the introduction of ChatGPT, with a one standard deviation increase in exposure associated with a £106 increase in average compensation, although the overall estimate is only marginally significant.

4.2 Job Posting Dynamics

4.2.1 Firm-Level Postings

While employment stocks adjust slowly, job postings provide a real-time window into firms’ evolving labor demand. We therefore examine how firms’ hiring intentions respond to LLM exposure by analyzing whether exposed firms post fewer vacancies overall and whether they adjust the composition of roles they seek to fill. This posting data captures the extensive margin of recruitment activity and reveals forward-looking decisions that precede the employment stock changes documented above. By tracking the same firms across both datasets, we can directly observe

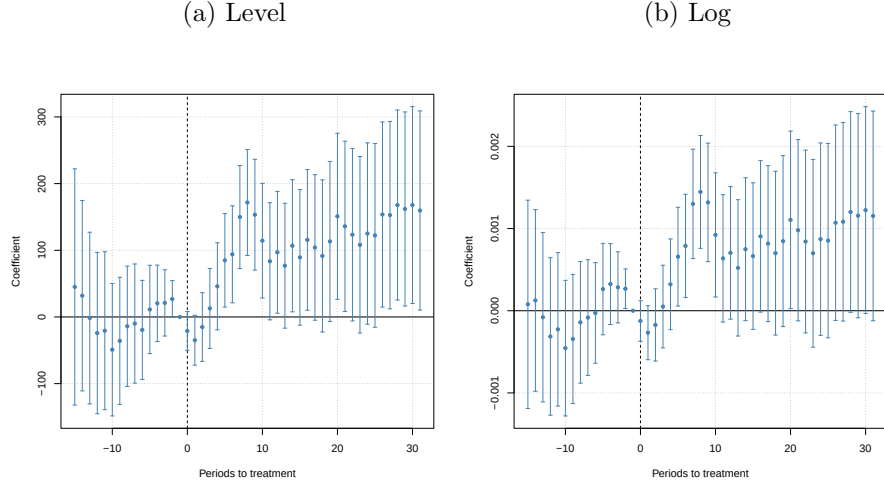
Figure 2: Effect of LLM exposure on firm-level seniority of jobs



Notes: This figure plots event study estimates of the effect of LLM exposure on the seniority composition of jobs within UK firms. Panel A shows effects on the total number of low-seniority positions (seniority levels 1–3), panel B on the total number of high-seniority positions (levels 4–7), and panel C on the fraction of jobs that are low-seniority. Coefficients are estimated from Equation (3). All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. The vertical line marks November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

how hiring intentions translate into gradual changes in employment.

Figure 3: Effect of LLM exposure on firm-level compensation



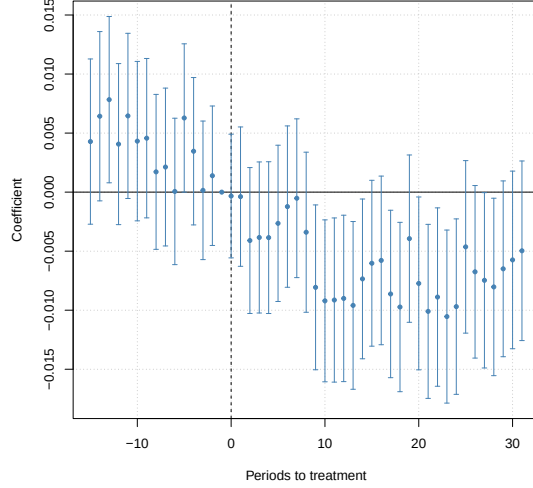
Notes: This figure plots event study estimates of the effect of LLM exposure on average compensation within UK firms. Compensation is defined as annual base salary plus bonuses. Panel A shows compensation in levels, Panel B in logs. Coefficients are estimated from Equation (3) and are shown relative to October 2022, the final pre-treatment month. All regressions include firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. The vertical line marks November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

Figure 4 presents event study estimates for the probability that exposed vs. less exposed firms post any job vacancy in a given month. The results show a change in hiring patterns following November 2022. Prior to this date, firms with different exposure levels did not have statistically significantly different trends in job postings.⁸ Following the release of ChatGPT, however, we observe that a one standard deviation increase in LLM exposure reduces firms' likelihood of posting new jobs by 1.0 percentage point ($p < 0.001$, see Table A1). This effect emerges gradually and becomes statistically significant after approximately 10 months, mirroring the timeline of employment stock adjustments.

Next, we decompose the aggregate hiring reduction by estimating occupation-specific effects across job categories. We estimate Equation (4) separately for 50 distinct occupational categories, with the outcome variable indicating whether a

⁸Although all pre-trend estimates are statistically insignificant, the point estimates exhibit a slight downward drift. If this reflects a differential pre-trend rather than sampling variation, the post-treatment estimates may overstate the true effect.

Figure 4: Effect of LLM exposure on firm-level job listings



Notes: This figure plots event study estimates of the effect of LLM exposure on the probability that a UK firm posts any job vacancy in a given month. Coefficients are estimated from Equation (3) with firm fixed effects and industry-by-period fixed effects. Standard errors are clustered at the firm level. The dependent variable is an indicator equal to one if the firm posted at least one vacancy in month t . The vertical line marks November 2022, when ChatGPT was publicly released. Shaded areas denote 95% confidence intervals.

firm posts any vacancy for that specific role in a given month. We continue to compare exposed vs. less exposed firms in the same time period.

Figure 5 presents these occupation-specific treatment effects, ordered by magnitude. The results reveal substantial heterogeneity in hiring adjustments across roles. Technical positions experience the sharpest declines: a one standard deviation increase in firm-level LLM exposure reduces the likelihood of posting software engineer positions by 1.1 percentage points ($p < 0.001$), QA tester positions by 1.0 percentage points ($p < 0.001$), and data analyst positions by 0.8 percentage points ($p < 0.001$). We find similar declines for other technical roles such as IT specialists, IT project managers, and infrastructure engineers. Creative and design roles also see significant reductions, with job postings for designer positions decreasing by 0.6 percentage points ($p < 0.001$) and for producers by 0.5 percentage points ($p < 0.001$). In contrast, sales representative positions show a statistically significant increase of 0.3 percentage points ($p < 0.001$). This pattern suggests that exposed firms are scaling back recruitment for technical and analytical roles where LLMs can augment or